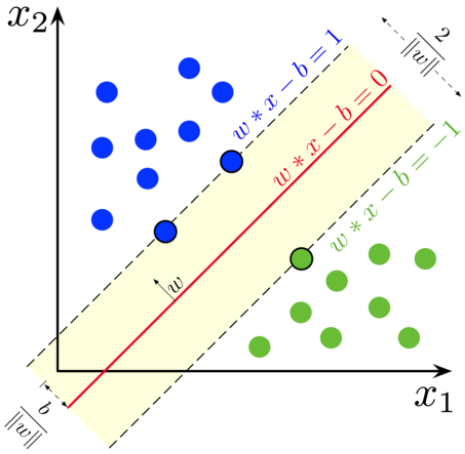
Lecture 10 - Support Vector Machines

SVM's are **hyperplanes** that maximize the distance to all classes. The **margin** should be as large as possible.



Linear SVM

The hyperplane is defined by:

$$w^T x - b = 0$$

The distance to both positive and negative classes should be as large as possible:

$$w^T x_+ + b \geq 1$$

$$w^T x_- + b \leq -1$$

The classifier for new instances is:

$$sgn(w^T x - b)$$

Non-Linear SVM

Soft-Margin: add slack variables to allow mistakes:

$$w^T x_+ + b \geq 1 - \zeta$$

$$w^T x_- + b \leq -1 + \zeta$$

Kernels:

1) Transform data to a higher-dimensional space where they hopefully will be linearly seperable

2) Fit the SVM in that higher-dimensional space

3) Transform linear SVM back to original space (not literally)

The optimization of SVM depends on the dot product, which means we need to find a kernel that optimizes the dot product.

$$K(x_i, x_j) = \Theta(x_i) \cdot \Theta(x_j)$$

Common kernels:

Polynomial Decision Surface: $K(x_1, x_2) = (x_1 x_2 + 1)^p$

Radial Basis Function / Gaussian Kernel:

$$K(x_1, x_2) = e^{-\frac{||x_1 - x_2||^2}{2\sigma^2}}$$

These kernels are guaranteed to converge, but can still overfit.

SVM Conclusions

SVM's are useful when:

- You want good performance

- You want off-the-shelf solutions with standard libraries

- You do not deal with too unstructured data

- You do not deal with too large datasets

- You are not rich but care for efficiency

- You do not want to deal with parameter optimization

Support Vector Regression (SVR)

SVR introduces flexibility in defining how much error is acceptable in our model. The optimization goal focuses on minimizing the coefficients instead of the total cost. For:

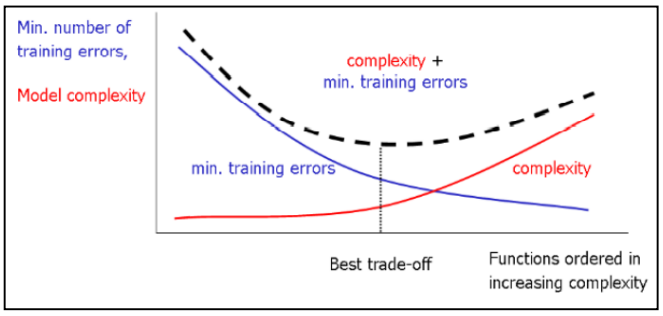
$$f(x) = < w, x > + b$$

Minimize $\frac{1}{2}||w||^2$ to:

$$y_i - < w, x_i > - b \leq \epsilon$$

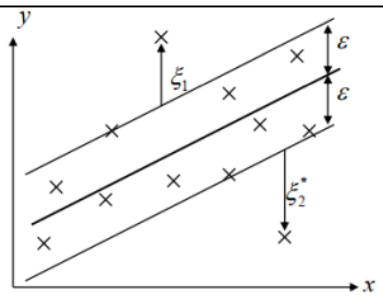
$$< w, x_i > + b - y_i \leq \epsilon$$

Training SVM:



Given training data

$$(\mathbf{x}_i, y_i) \quad i = 1, \ldots, m$$



Minimize

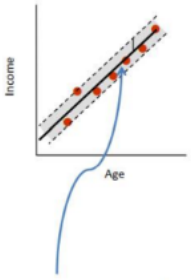
$$\frac{1}{2} \| \mathbf{w} \|^2 + C \sum_{i=1}^{m} (\xi_i + \xi_i^*)$$

Under constraints

$$\begin{cases} y_i - (\mathbf{w} \cdot \mathbf{x}_i) - b \leq \varepsilon + \xi_i \\ (\mathbf{w} \cdot \mathbf{x}_i) + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0, i = 1, \ldots, m \end{cases}$$

- **Linear case**

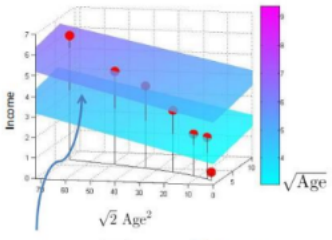
$f: age \rightarrow income$



$y_i = \mathbf{w}_1 \cdot \mathbf{x}_i + b$

- **Non-linear case**
  - Map data into a higher dimensional space, e.g.,

$f: (\sqrt{age}, \sqrt{2age^2}) \rightarrow income$



$y_i = \mathbf{w}_1 \sqrt{\mathbf{x}_i} + \mathbf{w}_2 \sqrt{2} \mathbf{x}_i^2 + b$